INTERACTIVE APPLET TO PRACTISE EXPONENTIAL FORM

Aim

To create an interactive GeoGebra applet for practising exponential form.

Procedure

Step 1: Open GeoGebra

- Open GeoGebra Classic and choose a view with the Input Bar and tools menu.

Step 2: Create variables for Question 1

- Enter these commands in the Input Bar:

```
base1=9
exp1=3
question1=base1^exp1
```

- This gives $9^3 = 729$.

Step 3: Display Question 1

- Choose the Text tool and display a question asking students to identify the base and exponent for the given number.
- For this click text option from tools menu and enable LaTeX Formula option for the mathematical expression. And enter question by below command

```
\text{Find the exponent and base of}:question1
```

- For a dynamic question, insert the current GeoGebra object question1 into the text object so it updates when their values change.

Step 4: Create input boxes for Question 1

- Create two answer variables:

```
ans1=0
ans2=0
```

- Create the Input Box from tools menu. In input dialog box set the caption to Base and link it to ans1. Similarly create another input box set the caption to Exponent and link it to ans2.

Step 5: Create the first Check button

- Create Boolean variable:

```
check1=false
```

- Create a button with caption Check. In script area, enter:

```
SetValue(check1,true)
```

- Click OK.

Step 6: Add feedback for Question 1

- Enter the following command in the Input Bar to display feedback only after the Check button is clicked:

```
feedback1=If(check1,If(ans1^ans2==question1," correct!" ," Wrong!" )," " )
```

- The entered base and exponent are correct when ans1^ans2 equals question1.

Step 7: Create variables for Question 2

- Enter these commands:

```
Base2=10
Exp2=3
question2=base2^exp2
```

- This gives $10^3 = 1000$.

Step 8: Display Question 2

- Choose the Text tool and display a question asking students to identify the base and exponent for the given number.
- For this click text option from tools menu and enable LaTeX Formula option for the mathematical expression. And enter question by below command

```
\text{Find the exponent of the number}:question2\times\text{with base}base2
```

- For a dynamic question, insert the current GeoGebra object question2 and base2 into the text object so it updates when their values change
- Click OK

Step 9: Create the answer input for Question 2

- Create answer variables:

```
ans3=0
```

- Create the Input Box from tools menu. In input dialog box set the caption to Exponent and link it to ans3.

Step 10: Create the second Check button

- Create Boolean variable:

```
Check2=false
```

- Create a button with caption Check. In script area, enter:

```
SetValue(check2,true)
```

- Click OK.

Step 11: Add feedback for Question 2

- Enter the following command in the Input Bar to display feedback only after the Check button is clicked:

```
feedback2=If(check2,If(ans3==exp2," correct!"," Wrong!"),"")
```

- If ans3 equals exp2, the applet displays Correct! Otherwise, it displays Wrong! Feedback remains blank until Check is clicked.

Step 12: Create the Next Question button

- Create a button captioned Next Question. Open its On Click scripting area and paste this complete script. The random ranges below are example choices for positive integer practice questions.

```
SetValue(base1,RandomBetween(1,10))
SetValue(exp1,RandomBetween(1,10))
SetValue(base2,RandomBetween(1,10))
SetValue(base1,RandomBetween(1,10))
SetValue(ans1,0)
SetValue(ans2,0)
SetValue(ans3,0)
SetValue(check1,false)
SetValue(check2,false)
```

- Click OK.
- The button generates new values, resets the answer inputs to zero, and clears feedback by resetting check1 and check2 to false. Make sure the displayed question text is dynamic and linked to the current GeoGebra objects.

Result

An interactive GeoGebra applet is created with two types of exponential-form questions, input boxes, Check buttons, feedback messages, and a Next Question button.

Find the exponential of this number 81 with base 3

Exponent 4 Check

correct

Find the exponential form of the number 9

base 3 exponent 2 Check

correct

New questions